

7

Occupational Medicine and Vocational Rehabilitation

ADAM SABY, MAURO ZAPPATERRA, AND DIXIE ARAGAKI

SUMMARY PEARLS

- Among a broad range of physical and psychological factors, employees who “hardly ever” enjoy their job were most likely to report a back injury.
- For conditions that require relative rest, time-based activity restrictions are easier for patients to follow and for employers to accommodate.
- When assessing causation of a work injury, use the employee’s occupational mechanism of action to differentiate occupational diagnoses from nonoccupational diagnoses.
- Workers’ compensation treatment guidelines ensure injured workers receive standardized evidence-based care that aims for optimal recovery and timely return to work while allowing flexibility for individualized treatment when necessary.
- Effective return-to-work interventions for injured workers involve early coordination of care, workplace modifications, therapeutic education, and a multidisciplinary approach to support sustained workability and optimal recovery.

Introduction

Occupational medicine and vocational rehabilitation are two distinct yet interrelated health care disciplines, with occupational medicine focusing on optimizing the health of the worker and vocational rehabilitation aiming to maximize work participation. Both incorporate a holistic, multimodal approach to identify and account for specific impairments, address activity limitations, and enable job participation. These disciplines accomplish this through policies, procedures, programs, or adapted interactions with the workplace environment from the level of the affected worker to the system at large. Through the lens of a psychiatrist, these fields of medicine share a familiar focus on patient-centered care using an interprofessional team to promote wellness, prevent and treat injuries, restore meaningful function, and adapt the environment to the working individual. The goals of occupational medicine and vocational rehabilitation are to preserve productivity and job performance respectful of any limitations using prevention, protection, rehabilitative treatment, adaptive equipment, and compensatory strategies.

Occupational Medicine

Occupational medicine (also known as occupational health, occupational and environmental medicine, and [formerly] industrial medicine) has evolved to encompass the clinical practice of assessing, treating, and preventing health disorders arising from or in association with job performance. It is the intersection of

medicine, employee health, research, and advocacy for maintaining a healthy workforce with maximal productivity and safety in the work environment. From a biopsychosocial perspective, health includes a person’s complete physical, mental, and social well-being. A worker’s fitness is not defined as just the absence of disease but described in terms of bodily function and structure, activity limitations, and/or environmental factors.⁶² The modern workplace can potentially entail any or all of the following elements: physical, chemical, biologic, social, recreational, spiritual, and psychological. Therefore the broad goals of occupational and environmental medicine are promoting workers’ health, improving the work environment for wellness, and developing a culture that supports safety. Such a “safety culture” should foster a positive social climate to enhance team productivity without injury at work and must be reflected in leadership direction, management priorities, employee attitudes, policies, procedures, and measurable reported outcomes. For example, during the COVID-19 pandemic, the spread of community illness was linked to workplace outbreaks, prompting widespread adaptations in public health, employee safety, surveillance practices, and business policies.²⁸

Brief History of Occupational Medicine

Hippocrates described lead poisoning in miners during the fourth century BCE, but the first book credited with formally introducing the specialty was an Italian text, *De Morbis Artificum Diatriba* (Diseases of Workers), by Dr. Bernardino Ramazzini in 1700. The

world was experiencing the industrial revolution with new bursts of productivity and innovation. With the paradigm shift from small-scale craftsmanship to mass-production factories, society and culture were subject to many changes, some beneficial and others burdensome. Ramazzini was a pioneer in recommending periods of rest, exercise, posture awareness, and hygiene to prevent work-related illnesses. There was an international movement to protect workers through age limits, work hour restrictions, safer building conditions, and safety standards to reduce hazardous exposures and risks of injury. In Germany, Chancellor Otto von Bismarck proposed the first modern system of workers' compensation insurance in 1884 (Workers' Accident Insurance) and advocated for an Old Age and Disability Insurance Law of 1889 with coverage of all worker categories, including industrial, agrarian, artisans, and servants.

The industrial revolution in the United States catalyzed increased building and manufacturing but burdened workers with higher expectations on pace, hours, and overall productivity (all of which are risk factors for musculoskeletal injuries). In the early 1900s, it was common for Americans to work 10-hour days, 6 days a week with little recourse to negotiate adequate compensation or safe working conditions. Unsurprisingly, work-related accidents and deaths were higher before the labor movement helped to bring advocacy and policies to protect workers. During the early 20th century, workers' compensation insurance rose in popularity due to increased awareness of dangerous working conditions and the need to offset the devastating financial impacts of work-related injury or illness for workers and families. Workers' compensation insurance is the oldest social insurance program in the United States, now mandated for most businesses in all states except Texas (where it is optional). It is commonly based on a no-fault system, meaning that an injured worker can secure cash benefits and medical care without having to demonstrate that the employer or another party was negligent or caused an on-the-job incident.⁴⁹ Workers' compensation insurance is a key deterrent in reducing workplace injury litigation. Over time, demand grew for physicians to help address the negative physical, environmental, and psychosocial impacts as well as evaluate, treat, and rehabilitate workers at the earliest opportunity. The outdated narrow term *industrial medicine*, used before 1960, was replaced by *occupational medicine* to incorporate broader elements of public health, environmental exposures, toxicology, and a wide variety of vocational injuries and disorders. Notably, psychiatrists possess the clinical skills, holistic viewpoint, and team-oriented perspective to be uniquely effective in this field with their additional capacity to focus on occupational rehabilitation.

Epidemiology of Work-Related Fatality, Injury, and Illness

Between 1912 and 2022, the rate of unintentional work deaths dropped from 21 to 3.7 per 100,000 American workers. A total of 5,486 fatal work injuries were recorded in 2022 according to the US Bureau of Labor Statistics.⁵⁴ The highest number of workplace fatalities in 2022 was transportation related (38%), and the second highest was in construction and extraction workers (19%). Occupations with the highest fatality rate (23.5 per 10,000 full-time equivalent [FTE] workers) were in the farming, fishing, and forestry industries. Notably, unintentional injuries of all causes (occupational and nonoccupational) have been the third leading cause of death in the United States between 2018 and 2022, trailing heart disease and cancer, followed by COVID-19, cerebrovascular disease, and chronic lower respiratory diseases.³⁴ Climate

change also contributed to increasing work-related fatalities in 2022 due to temperature extremes, specifically environmental heat. Private industry employers reported 2.8 million nonfatal workplace injuries and illnesses in 2022 at a rate of 2.3 cases per 100 FTE workers.⁵⁵ The increase was driven by a modest rise in musculoskeletal injuries and a sizable increase in respiratory illnesses secondary to COVID-19.

The most common reason for missed work in 2021 and 2022 was "exposure to harmful substances or environments," a category that includes cases of COVID-19. The next two most prevalent causes of injury and illness were "overexertion and bodily reaction" and "slips, trips and falls." These top three categories accounted for more than 75% of all nonfatal injuries and illnesses leading to days away from work.³⁸ Incidents related to overexertion involve lifting, carrying, throwing, and twisting type motions and can lead to work-related musculoskeletal disorders. The National Safety Council has focused on the dangers that fatigue brings to the work environment because research reveals that workers with sleep disorders are 1.6 times more likely to be injured, and 13% of job-related injuries are attributed to workers with fatigue.^{32,46} Fatigue also impacts driver safety. Nonfatal occupational injuries are common in nursing and residential care facilities, veterinary services, police and fire protection services, manufacturing, and construction. Musculoskeletal sprains/strains and pain/soreness are the most reported conditions. However, other common occupational ailments are related to toxin exposure (e.g., pesticides, lead), hearing loss, heat/cold/radiation exposure, mental illness, cancer (up to 5%–10% of malignancies are considered work related), infections, and respiratory diseases (e.g., an estimated 15% of new-onset asthma is work related).³⁰

Injury incidence differs by sex, age, and race. Males comprise most occupational fatalities. Whereas most fatal work injuries occurred in groups between 25 and 65 years old, 14% of the fatally injured workers in 2022 were age 65 years or older.⁵⁴ Workers who identified as White or Hispanic/Latino had higher rates of death compared to Black, Asian, or Indigenous workers.⁵⁴ In addition to loss of life and livelihood, workplace injuries and illnesses include direct and indirect costs. Direct costs include workers' compensation payments, medical expenses, and costs for legal services. Indirect costs include training replacement employees, accident investigation, implementation of corrective measures, lost productivity, repairs of damaged equipment/property, lower employee morale, and absenteeism.

Safety in the Workplace

Safety Regulation

The Department of Labor (DOL) addresses the millions of injuries and work-related deaths in the United States each year. Under the DOL, the Occupational Safety and Health Administration (OSHA) is the regulatory body tasked with creating and enforcing occupational safety standards for private and public sector employers nationwide. OSHA's general duty clause states each employer shall provide "a place of employment which are free from recognized hazards that are causing or are likely to cause death or serious physical harm to his employees." The National Institute for Occupational Safety (NIOSH) is a research body tasked with creating guidance for enforcement agencies, such as OSHA, on how to best prevent and address occupational injury.

To monitor safety, OSHA requires employers to keep records of all work-related reportable injuries, including any injury needing medical management beyond first aid (e.g., x-rays, prescription medications). OSHA also conducts planned inspections for hazardous industries or when there is a specific event, such as an occupational catastrophe, severe injury (e.g., work-related fatality), employee complaint, or spike in reportable injuries. To improve safety, OSHA provides training for public and private sector safety officials on how to comply with safety regulations.

Safety Strategies

Safety specialists, including hired consultants or those housed within a company's environmental health and safety (EH&S) department, help employers adhere to state and federal regulations. Often accredited by the Board of Certified Safety Professionals, these specialists develop strategic plans to lower an employer's injuries as compared with industry standards. Examples of successful injury prevention programs featured in an OSHA White Paper show top-level leadership, employee participation, and a "find and fix" strategy are effective.⁴¹

The hierarchy of hazard control is an organizational system used by OSHA and safety specialists to detail the stepwise approach in implementing safety measures. The five tiers of intervention proposed from most to least effective are elimination, substitution, engineering controls, administrative controls, and personal protective equipment (Fig. 7.1).³⁷ Elimination and substitution are concerned with removing or replacing hazards, but they are best instituted when first designing the work project. When these top-level strategies are not feasible, engineering controls help to isolate workers from the hazard. For example, a laboratory biosafety cabinet with unidirectional airflow can separate workers from hazardous gases or liquids, especially in the event of a spill. The next safety tier is administrative controls, which involve changes in workplace practice and policies. Examples include reducing the time spent with hazardous materials by rotating workers through workstations or educating workers to

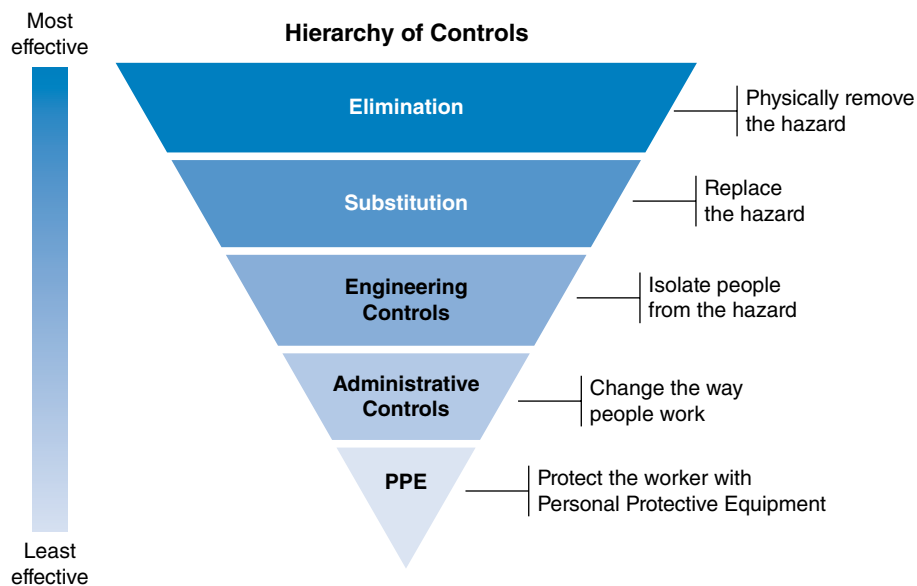
recognize early symptoms of occupational injury. The last and least effective tier of safety is personal protective equipment (PPE). PPE provides an immediate wearable barrier from the hazard and includes respirators, ear plugs, safety goggles, hard hats, flame-retardant aprons, bulletproof vests, and gloves.

Ergonomics is "the science of fitting workplace conditions and job demands to the capability of the working population."³⁶ The goal of ergonomics is to prevent injuries and disorders associated with uncomfortable postures or cumulative trauma from repeated tasks. A workplace ergonomics program aims to reduce work-related musculoskeletal disorders by mitigating awkward positioning, repetition, significant force, mechanical compression, vibration, temperature extremes, glare, inadequate lighting, and duration of exposure. Examples of ergonomic interventions include reducing the immediate physical effort to transport materials (e.g., via mechanical lift) or reducing the repetitive physical effort while using a workstation (e.g., via computer desktop ergonomics) (Box 7.1; Fig. 7.2).^{40,45}

Workers With a Disability

Preexisting Disability

The Americans with Disabilities Act (ADA) of 1990 was established to provide greater employment opportunities to those with disabilities by preventing discrimination based on a physical or mental impairment. The ADA, as enforced by the Equal Employment Opportunity Commission, requires antidiscrimination measures be followed by all public and private sector employers with 15 or more employees.⁴⁷ The ADA prohibits employers from asking applicants medical questions during the interview process or prior to making a job offer. Once the applicant accepts an offer, the employer may require a preplacement physical to assess if the new employee can perform the job's essential functions. A preplacement physical yields valuable information about a new employee's ability to accomplish essential job tasks. Employees with a mismatch between their physical capabilities and their job tasks



• **Fig. 7.1** National Institute for Occupational Safety and Health (NIOSH) five-tiered hierarchy of controls for injury prevention. PPE, Personal protective equipment. (From Centers for Disease Control and Prevention, National Institute for Occupational Safety and Health. *Hierarchy of Controls*. <https://www.cdc.gov/niosh/topics/hierarchy/default.html>.)

• BOX 7.1 Workstation Ergonomic Recommendations to Promote Neutral Body Positioning and Reduce Work-Related Musculoskeletal Disorders

Recommendations to maintain neutral body postures while using a computer workstation:

- *Hands, wrists, and forearms* are straight and parallel to the floor.
- *Head* is level and balanced, in line with the *torso*.
- *Eyes* are in line with top of screen. (Bifocal lens users can lower the primary display 30 to 45 degrees below eye level to avoid hyperextension of the *neck*.)
- *Shoulders* are relaxed and *upper arms* hang naturally at the side of the body.
- *Elbows* stay close to the body and are flexed between 90 and 120 degrees. Keep frequently used items within reach to limit leaning, bending, or twisting at the *waist*.
- *Feet* are fully supported by the floor or a footrest.
- *Back* has appropriate lumbar support when sitting vertical or leaning back slightly.
- *Thighs and hips* are supported and generally parallel to the floor.
- *Knees* are about the same height as the hips with the *feet* slightly forward.

Data from Occupational Safety and Health Administration. *Computer Workstation Positions*. <https://www.osha.gov/SLTC/etools/computerworkstations/positions.html>; Rempel DM, Janowitz IL. Ergonomics and the prevention of occupational injuries. In: *Current Occupational and Environmental Medicine*. 5th ed. McGraw-Hill Education; 2014:197-222.



• **Fig. 7.2** A comfortable working posture in which the joints are naturally aligned. Working with the body in a neutral position reduces stress and strain on the muscles, tendons, and skeletal system and reduces the risk of developing a musculoskeletal disorder. (From US Department of Labor, Occupational Safety and Health Administration. *Computer Workstations eTool*. <https://www.osha.gov/SLTC/etools/computerworkstations/>.)

will have increased occupational injuries.¹⁹ If new employees pass the preplacement physical requirements, according to the ADA, employers must provide reasonable accommodations for preexisting disabilities. Accommodations can be physical (e.g., a wheelchair access ramp) or administrative (e.g., earlier shift start time). Only if accommodations create “undue hardship,” by way of

significant difficulty or expense, can an employer rescind the job offer.

Preplacement and Medical Surveillance Exams

Physical medicine and rehabilitation (PM&R) physicians can help prevent injuries in the workplace and accommodate employees with disabilities through the preplacement physical exam. The primary goal of the preplacement physical is to assess if the employee can safely perform the job’s essential functions. These essential functions may include lifting objects up to a maximum weight (e.g., 50 pounds) or identifying traffic light colors when driving (e.g., red vs. green). As a result, the physician must focus the history and physical to assess the employee’s ability to accomplish specific work tasks. If the employee is unable to complete portions of the exam, the physician must communicate these limitations to the employer.

In addition to the physical assessment, the physician investigates for preexisting disabilities and the need for specific work restrictions. Certain medical diagnoses, such as severe pulmonary hypertension, may require clarification from the employee’s primary care physician about the need for medical restrictions (e.g., limit prolonged walking). If the medical restrictions for a preexisting disability do not interfere with the job’s essential functions, the physician relays the information so the employer can provide reasonable accommodations.

A third component of the preplacement physical is identifying opportunities for occupational injury prevention. Occupations can expose employees to endemic diseases (e.g., malaria for researchers traveling to Tanzania) or have unique exposures (e.g., human blood exposures in a hospital setting). In many cases, physicians can minimize occupational hazards by offering chemoprophylaxis (e.g., anti-malarials) or vaccination (e.g., hepatitis B vaccination) during the preplacement physical. Hazards that require regular vaccine boosters (e.g., rabies) or screening (e.g., tuberculosis in a K-12 school setting) can spur the creation of a routine medical surveillance program.

A medical surveillance program involves recurring health assessments to prevent and monitor for occupational injury. These assessments may involve a health questionnaire to review changes to the employee’s medical status (e.g., new immunocompromised condition) or job roles (e.g., a pharmacist takes on a new role that requires compounding chemotherapeutic medications). The assessment may require a focused physical exam and diagnostic studies to screen for early signs of illness (e.g., annual chest x-rays for asbestosis in construction workers). The Department of Transportation (DOT) medical exam is a nationwide example of a medical surveillance program that assesses the health of employees who drive commercial motor vehicles. The DOT exam monitors for various health conditions, including those that may cause sudden incapacity (e.g., epilepsy), which may affect the safe operation of a commercial motor vehicle.

Occupational Injury and Illness

Despite best efforts to promote occupational safety, injuries occur. Under the workers’ compensation insurance system, an employee fills out a claim form explaining the injurious episode. The employee notes the date and mechanism of injury as well as the injured body parts (e.g., left shoulder, low back, left knee). At this point, the physician obtains a more detailed medical history and performs a physical exam. As part of the assessment and plan, the physician must determine causation of the injury as occupational versus nonoccupational.

A basic standard is that to be considered occupational and eligible for workers' compensation benefits, the injury must arise out of employment (AOE) and occur during the course of employment (COE). In other words, the injury must have occurred on the job and within the scope of the job's duties. For example, a mail carrier who falls on the sidewalk while delivering mail for the employer meets AOE/COE criteria. On the other hand, a mail carrier who falls on the sidewalk while visiting a restaurant during a lunchbreak does not meet AOE/COE and, as a result, is not eligible for workers' compensation benefits.

While insurance claims adjusters can help determine AOE/COE, only a physician can determine which diagnoses are occupational versus nonoccupational. This process, called apportionment, helps determine what diagnoses are covered for treatment in the occupational medicine setting. A PM&R physician has a unique skill set in determining apportionment because of familiarity with the biomechanics of trauma. The employee's mechanism of injury should directly cause the occupational diagnosis. Other diagnoses not directly caused by the cited mechanism of injury are considered preexisting, whether known or unknown to the employee. For example, a school janitor sweeping the floors hits the anterior aspect of the knee against a classroom desk. After a thorough history and physical, the employee's diagnoses include a patellar contusion and tibiofemoral osteoarthritis. By reviewing the mechanism of injury cited (i.e., a low-amplitude blunt trauma to the anterior knee), the PM&R physician assesses that the patellar contusion is occupational, but the tibiofemoral osteoarthritis is a preexisting nonoccupational condition. As a result, workers' compensation benefits cover treatment for the occupational diagnoses (e.g., patellar contusion) while preexisting diagnoses (e.g., osteoarthritis) are deferred to the employee's primary care physician for management.

A common series of injuries in the workplace are cumulative trauma disorders. The mechanism of injury cited typically includes weeks to months of repetitive work tasks. One of the most common cumulative trauma disorders, often resulting from repetitive keyboarding and mousing at computer workstations, is carpal tunnel syndrome (CTS).¹⁴ A pooled study of workers from various industries found that peak hand force and time spent performing forceful hand exertion are associated with an increased risk of CTS.²⁰ Though increased computer mouse use and exposure to significant vibration (e.g., operating a jackhammer) have been implicated as risks,¹ there are multiple nonoccupational factors that also predict CTS, including obesity, pregnancy, smoking, and older age.³³ While ergonomics can minimize physical discomfort with equipment and workstation design, addressing relevant nonoccupational conditions is necessary for healing.⁵³

Determining Temporary Disability

In addition to diagnosing and treating an employee's occupational condition, a PM&R physician has a key role in creating the employee's activity prescription (i.e., Activity Rx). Through an understanding of neuromusculoskeletal physiology, the physician writes a set of biomechanical restrictions an employee can follow (e.g., limit walking for an ankle sprain) to ensure optimal healing. Successful biomechanical restrictions can be applied to the patient in any setting, including work, home, during travel, and more. As a result, a range of activities the patient can safely partake in is also implied (e.g., answering phone calls or using a computer). An effective Activity Rx prevents unnecessary total disability and promotes functional independence.

Time-based restrictions (e.g., limit walking to 15 minutes per hour) are especially useful when a medical condition requires relative rest as opposed to total rest. Though granting an employee time off from work (i.e., temporary total disability) can create patient satisfaction, it creates an unrealistic expectation about healing as an all-or-none phenomenon. Setting expectations about advancing activity to coincide with rehabilitation treatment helps employees achieve their functional goals in a stepwise process.

Along with specifying restrictions, an Activity Rx should be simple enough for the employee to follow and for the employer to accommodate. With information in the Activity Rx, a vocational rehabilitation or disability management specialist helps to return the employee to a suitable working environment. The objective is limiting the employee's temporary total disability because the longer the employee is out of work, the more likely the employee will not return to productive employment.¹¹ Also, early mobility with multiple musculoskeletal conditions enhances recovery. For example, workers with subacute low back pain who received a recommendation to "stay active" returned to work significantly earlier than those who received standard primary medical care.³²

A physician should learn from the employer's disability specialist about what specific activity restrictions hinder or help the employer's ability to accommodate. The best accommodation is keeping an employee in the same working environment by placement on a part-time or rotational work status. If this is not feasible, the next form of accommodation is finding alternative work, sometimes in a different department, to keep the employee tied to the general workplace. Vocational rehabilitation methods using graded work conditioning and work hardening increase the likelihood of returning to full employment.⁴⁸

Treating Temporary Disability

Work-related musculoskeletal disorders are the most common occupational medicine conditions. Of those, low back injuries are the leading cause of workplace disorders and lost workdays.²² Back pain is strongly influenced by an employee's self-perception of having poor general health and feeling exhausted.¹⁸ While not the cause of an employee's diagnosis, an employee's perception of work has a strong influence on prognosis. A pivotal study evaluating over 3,000 Boeing aircraft employees found the strongest predictor for filing a low back workers' compensation claim was low job satisfaction.³ Compared to patients without workers' compensation claims, those with claims are more likely to have worse functional status and receive disability compensation at 4 years postinjury.² Other factors that prolong disability include psychological issues, such as depressive symptoms, catastrophizing, and low self-confidence.⁴²

As such, interventions to prevent and treat disability must focus on measures beyond tissue pathology. A holistic approach that addresses a patient's overall mental and physical health leads to better disability outcomes. Return to work is based on biopsychosocial elements such as recovery beliefs and pain behaviors. The patient's expectation of recovery is one of the most consistent predictors of returning to work from any health condition.⁷ On the other hand, unrealistic treatment expectations, such as those with surgical outcomes in low back pain, cause decreased satisfaction, reduced treatment compliance, and lower return to work.^{7,43} Managing recovery expectations with a patient helps identify those at risk for delayed recovery.

Work Is Therapy

The ability to work is a major predictor of a person's self-worth and provides people with a sense of purpose. An individual's employment status is not only meaningful to one's financial security but is also linked to an individual's long-term health. It has been shown that there is a strong relationship between unemployment and all-cause mortality, including suicide and cardiovascular disease.^{25,39} Individuals who are unable to return to work because of a disability may experience increased physical pain, deconditioning, anxiety, depression, social isolation, and financial hardship. Total disability decreases quality of life and increases health care utilization. Within limits, working is therapeutic. A PM&R physician who understands how employment enhances well-being can better align a patient's expectations with rehabilitation goals. A systematic review assessing lumbar surgery for radiculopathy found patients were more likely to return to work when they were working preoperatively or had shorter duration of preoperative symptoms.²⁶ Furthermore, longer preoperative disability and depression predicted longer postoperative disability.

Determining Permanent Disability

Employees reach maximum medical improvement (MMI) when they have received the standard medical workup and treatments used to manage their occupational condition. At this point, further treatment is not expected to have a significant impact on functional status. As a result, if employees cannot return to regular duty, their temporary partial disability is assessed as a permanent partial disability. In conversation with patients, the physician determines the restrictions for permanent partial disability to maximize patients' function in the workplace while minimizing flareups of their condition.

Once a physician dictates permanent restrictions, the employer must engage the employee in the interactive process to see if long-term accommodations can be made. Often, the ability to accommodate depends on the severity of the restriction, such that mild limitations (e.g., limit lifting to 50 pounds) are more likely to be accommodated by the employer than severe limitations (e.g., limit lifting to 5 pounds). If the permanent restrictions prevent an employee from completing essential job functions, then the employer can legally terminate the person's employment through medical separation. In this case, the employee may be eligible to receive educational funds for training in alternate occupations.

To better include workers with a permanent partial disability in the US workforce, there are federal regulations and incentives for employers. Section 503 of the Rehabilitation Act of 1973 prohibits federal contractors and subcontractors from discriminating against individuals with disabilities (IWDs). It also requires employers to take affirmative action to recruit, hire, and retain IWDs to make up 7% of their hired workforce. The ADA Amendments Act of 2008 provides job opportunities for IWDs and updates nondiscrimination provisions.⁵² Employers hiring workers with a new disability on vocational rehabilitation can be eligible for annual tax credits.⁵⁷

There are several agencies that provide vocational resources to IWDs. The federal government sponsors programs such as the Employer Assistance and Resource Network (EARN),¹² which helps IWDs to pursue careers. The Veteran's Administration (VA) provides a well-known vocational rehabilitation program to veterans with service-connected disabilities. The VA vocational program, which boasts an 89% satisfaction rate among participants,

offers job placement assistance, advises veterans on small business ventures, and provides apprenticeship opportunities.⁵⁶ Moreover, veterans who completed the program earned on average \$18,000 more in annual income than participants who did not. USAJobs is the federal government's official employment site that posts noncompetitive appointments for its agencies, such as the National Aeronautics and Space Administration (NASA), to employees with disabilities.⁵⁸ Nonprofit organizations, such as the National Industries for the Blind, also help individuals with specific disabilities achieve employment.³⁵

Vocational Rehabilitation

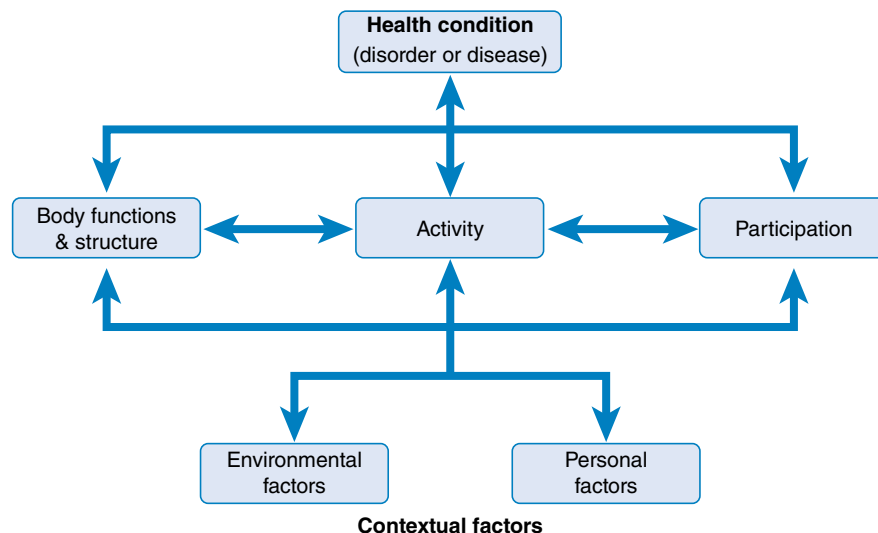
Vocational rehabilitation is a multidisciplinary approach to optimize work participation for employees with health-related impairments, limitations, or restrictions.¹³ Vocational rehabilitation benefits individuals who have a health condition, injury, illness, or disability that may impact their work participation. Vocational rehabilitation practitioners practice the integrative biopsychosocial approach of the World Health Organization's *International Classification of Functioning, Disability and Health* (ICF) and assess the variety of factors that can lead to work disability at the individual and societal levels. Vocational rehabilitation can improve return to work, increase functioning, and reduce pain.²⁹ Specific interventions can be used to address various impairments: Exercise can address physical disability, cognitive behavioral therapy can address psychosocial issues, graduated work simulation activities can assist job training, and supportive counseling can address labor, employer, insurance, and social issues.

Biopsychosocial Approach to Work Disability

Given today's demand for worker productivity and workplace wellness, vocational rehabilitation is a key mitigating measure for workplace disability. From a global perspective, Article 23 of the United Nation's Universal Declaration of Human Rights specifies that all people have the right to work, and the International Labor Organization ensures that all people have decent work.^{24,61} A common message from these international bodies is that an emphasis on overall health can be facilitated by enhancing an individual's ability and capacity to work. When work disability arises, it is key that appropriate steps are taken to mitigate its negative effects on workers, their families, and society.

Vocational rehabilitation programs aim to improve productivity of individual workers and their sense of social security, well-being, and personal fulfillment. Vocational rehabilitation programs are most effective when they utilize a holistic worker-centered approach.¹⁵ This approach is based on a biopsychosocial model, such as the ICF (Fig. 7.3), where a particular health condition, injury, or illness can lead to work disability. The ICF is a comprehensive framework for organizing information on functioning and disability. It conceptualizes functioning as a "dynamic interaction between a person's health condition, environmental factors, and personal factors." The ICF integrates both the medical and social models into a biopsychosocial synthesis. This approach acknowledges the roles of both health conditions and environmental factors in disability.

This biopsychosocial model is particularly effective for understanding work disability. A particular health condition, injury, or illness can lead to work disability, which may be associated with physical, mental, cognitive, sensory (vision and hearing), emotional, or psychological health conditions. Work disability can



• **Fig. 7.3** International Classification of Functioning, Disability, and Health (ICF). (From World Health Organization. *International Classification of Functioning and Disability and Health*. WHO; n.d.)

manifest as an impairment of body function (e.g., weakness) and body structure (e.g., muscle), limitation of activity (e.g., limited lifting ability), and/or restriction of participation (e.g., restricted employment). The level and extent of work disability is influenced by contextual factors such as the environment (e.g., workplace support and accommodation) and personal factors (e.g., motivation and coping). The ICF's inclusive approach recognizes various forms of work, including paid jobs and volunteer roles with comparable tasks.

The ICF employs a biopsychosocial approach to describe functioning and disability as umbrella terms that encompass the positive and negative aspects of health from the multiple biologic, individual, and social perspectives. The ICF framework is divided into two main parts: (1) functioning and disability and (2) contextual factors. Each part is further subdivided as follows:

1. Functioning and disability
 - Body functions and structures
 - Activities and participation
2. Contextual factors
 - Environmental
 - Personal

The ICF (see Fig. 7.3) highlights that functioning and disability are outcomes of interactions between health conditions (diseases, disorders, injuries) and contextual factors. Human functioning is classified at three levels: body or body part, the whole person, and the whole person in a social context. Disability involves dysfunction at one or more of these levels: impairments, activity limitations, and participation restrictions. Contextual factors include external environmental factors (e.g., social attitudes, architectural characteristics, legal and social structures, climate, and terrain) as well as internal personal factors (e.g., sex, age, coping styles, social background, education, profession, past and current experiences, overall behavior patterns, and character).

Vocational Rehabilitation Assessment and Evaluation

A comprehensive vocational assessment is critical in the development of a vocation-focused plan of care. Comprehensive assessments

involve the description and evaluation of nonmodifiable aspects (e.g., sociodemographic), health condition–related factors, environmental factors, functional abilities, and restrictions. All these components are vital to determine and optimize work-related capacities for part-time, full-time, light, modified duty, or self-employment settings. Currently, two major approaches to assess work functioning exist: patient-reported instruments and clinician-administered testing of abilities. Both approaches, with their respective advantages and disadvantages, complement each other to provide a whole picture of the worker.

Objective tests are standardized to measure tangible outcomes and are recognized by legal bodies but are usually labor intensive. One common example of objective testing is the functional capacity evaluation (FCE). Conversely, information from self-reported instruments is easy and fast to obtain but reflects the subjective evaluation of the patient, including perceived environmental barriers. In addition, the patient requires an appropriate literacy level for proper responding. An example of a questionnaire that is used in the vocational rehabilitation setting is the Work Rehabilitation Questionnaire (WORQ).¹⁵ One advantage of WORQ is that it was developed based on the ICF model and intended for generic needs in vocational rehabilitation. Despite differences between the patient-reported and clinician-administered testing, recent studies conclude that integrating the two approaches might complement each other in terms of time, costs, and information gained.

Another aspect that is essential to a vocational rehabilitation assessment is determining the educational needs of the worker and the type of job in which the worker will be engaging or reengaging. Having such knowledge will help guide the vocational rehabilitation practitioner in managing or referring services that would develop and/or enhance skills essential to employment, such as reading, writing, and problem solving. Investigating the relationship between the worker and the employer reveals employment potential and support of the employer for any workplace accommodation.

Workers' Compensation Guidelines

Workers' compensation treatment guidelines are established to ensure injured workers receive appropriate and effective medical

care following a work-related injury or illness. These guidelines are developed based on evidence-based medicine, best practices, and input from health care professionals. They aim to standardize treatment protocols to achieve optimal outcomes for injured workers. Treatment guidelines are typically based on best practices in the field of occupational medicine and rehabilitation. They may be regularly updated to reflect advances in medical knowledge and changes in treatment standards. While specific treatment guidelines can vary depending on the jurisdiction and the nature of the injury, there are some common elements: evaluation and diagnosis, treatment modalities recommended, duration and frequency of treatment, criteria for referral, return-to-work recommendations, follow-up, monitoring, and reassessment.

Treatment guidelines often outline the process for evaluating and diagnosing work-related injuries or illnesses. This may involve initial assessments by health care providers, diagnostic tests (e.g., x-rays, magnetic resonance imaging), and consultations with specialists as needed to determine the extent and nature of the injury. They often include recommendations for facilitating the injured worker's return to work in a safe and timely manner and expected days off from work. This may involve gradual increases in work activities, modifications to job duties or work environments, and coordination with employers to ensure a smooth transition back to work.

The guidelines also specify the types of treatments and interventions that are considered appropriate. This may include medications, physical therapy, chiropractic care, acupuncture, occupational therapy, and other rehabilitative services aimed at relieving pain, restoring function, and facilitating recovery. The guidelines help to provide recommendations regarding the duration and frequency of various interventions for diverse types of injuries. This helps ensure that injured workers receive timely and appropriate care without unnecessary delays or overutilization of services.

Finally, the guidelines typically emphasize the importance of ongoing monitoring and reassessment of the injured worker's progress throughout the treatment process. This allows health care providers to adjust treatment plans as needed based on the individual's response to treatment and any changes in the condition. By following workers' compensation treatment guidelines, health care providers can ensure that injured workers receive appropriate and effective care focused on restoring function, improving outcomes, and promoting a timely return to work. These guidelines ensure consistency and fairness in the provision of medical services to injured workers across different health care settings.

Using guidelines to authorize treatment in workers' compensation cases is crucial for several reasons. First, treatment guidelines provide standardized protocols for evaluating and managing work-related injuries or illnesses, ensuring consistency and uniformity in the care provided to injured workers, regardless of the health care setting or provider involved. Second, these guidelines are typically based on current medical evidence and best practices in the field, ensuring that injured workers receive treatments supported by scientific research and likely to produce the best outcomes. Third, treatment guidelines often include recommendations for cost-effective interventions that prioritize the efficient use of resources while delivering high-quality care and helping payers contain costs associated with workers' compensation claims. Additionally, guidelines provide clear criteria for determining when specific treatments are medically necessary and appropriate, streamlining the authorization process, reducing delays, and ensuring timely access to necessary care. They help ensure the

quality and appropriateness of care by establishing benchmarks and standards for evaluating treatment outcomes, allowing health care providers to demonstrate that they meet established standards of quality and effectiveness. In many jurisdictions, treatment guidelines allow payers to demonstrate they are fulfilling their legal obligations to provide appropriate medical benefits to injured workers.

Typically, physicians can use state guidelines to recommend treatment for injured workers. If state guidelines do not exist, then national guidelines, such as those found through the American College of Occupational and Environmental Medicine or Official Disability Guidelines (ODG), can be used. While adhering to treatment guidelines in workers' compensation cases is beneficial, there are situations where deviation from these guidelines may be necessary or even appropriate. Treatment guidelines may not fully account for the unique characteristics and needs of individual patients. Deviating from guidelines allows health care providers to tailor treatment plans to meet the specific circumstances of each injured worker, considering factors such as age, comorbidities, psychosocial factors, and preferences. Deviation from guidelines may be justified when there is compelling evidence to support alternative approaches not yet incorporated into the guidelines. Complex cases involving multiple comorbidities may also require deviation from guidelines to develop a comprehensive and individualized treatment plan. Furthermore, patients may have preferences or values that differ from the recommendations outlined in treatment guidelines, and health care providers should engage patients in shared decision making and respect their autonomy in determining their treatment preferences. In certain situations, deviation from treatment guidelines may be justified if the physician can demonstrate the alternative approach is medically necessary and reasonable under the circumstances. Health care providers should document their rationale to mitigate potential legal risks. While deviation from treatment guidelines should not be taken lightly, it is essential to recognize that guidelines serve as general recommendations rather than rigid mandates. Health care providers must balance adherence to guidelines with the need to provide individualized, patient-centered care that optimizes outcomes and respects patients' preferences and values.

Return-to-Work Interventions

Return to work following musculoskeletal injuries is a complex topic, and the outcomes are variable. Return to work is not solely dependent on the degree of injury and physical ability but is influenced by a range of patient-related factors and external circumstances that are all within the four domains of the ICF framework. Similarly, the outcomes for returning to work can range from returning to the previous job with no modifications, returning to the previous job with modifications, returning to the same workplace in a different capacity, or finding a new job altogether. The goal is for the injured worker to heal and be able to sustain work after the injury occurs, with the ideal outcome of lasting ability to work.

Early intervention with coordination of care and rehabilitation is integral to leading a person back to working life after an injury or illness. This may include workplace modifications, therapeutic education, increasing physical fitness, work conditioning, decreasing anxiety or depression, managing stress, and increasing self-efficacy.^{4,8} A review determined that the top five interventions for patients with occupational injuries include (1) acquiring skills; (2) health services, systems, and policies; (3) products and

technology for employment; (4) handling stress and other psychological demands; and (5) apprenticeship.⁵ The top 10 outcomes for measurement include (1) lucrative employment; (2) sensation of pain; (3) emotional functions; (4) handling stress and other psychological demands; (5) economic self-sufficiency; (6) muscle endurance functions; (7) exercise tolerance functions; (8) undertaking multiple tasks; (9) acquiring, keeping, and terminating a job; and (10) looking after one's health.⁵

Another recent systematic review aimed to determine the current state of therapeutic methods used in rehabilitation for individuals with back pain and to identify the most effective therapy for enhancing or restoring workability. Interventions that incorporate a physical therapy approach with a significant emphasis on exercise supervised by trained coaches are clinically effective in improving pain intensity, disability, and quality of life. In addition, a manual therapy program that includes stay-active therapy, stretching, and manual manipulation improved return-to-work rates. Additionally, the social interaction between the workplace and the employees is a crucial factor in work-related outcomes.²⁷

When workers are injured, simple strategies of workplace interventions, such as environmental modifications and schedule changes or modifications to duties and tasks, can help. In people with musculoskeletal disorders, workplace interventions reduced time to return to work and increased incidence of the return more than usual care.⁵⁹ Based on DOL data, employee accommodations are very cost effective, with over one-half of accommodations costing nothing to the employer.⁵³

A review looking at effectiveness of workplace interventions found interventions primarily in three domains: health focused, service coordination, and work modification. The health-focused interventions included graded activity, exercise, cognitive behavioral therapy, work hardening, and multicomponent interventions (a combination of medical assessment, physical therapy, psychology, and occupational therapy). It has been shown the physical conditioning programs for people with back and neck pain that incorporate a cognitive behavioral approach reduce sick leave when compared to usual care.⁴⁶ The service coordination interventions included communication between health care providers and workplace, developing return-to-work plans, education, and training. The work modification interventions involved modified duties, modified hours, and ergonomic or worksite changes. There is strong evidence that multidomain interventions from two of the three abovementioned domains reduced duration away from work, lowered costs, and improved work functioning.

For individual interventions, there is moderate evidence that work accommodations improved lost time from work. From a recent review, the most commonly reported accommodations included modifying job responsibilities, changing workplace policies, providing supportive personnel, implementing flexible scheduling, and offering assistive technology.⁶⁰ Therefore, for improved return to work, implementing graded activity with modified duties and implementing a multidomain intervention for patients with musculoskeletal and pain-related conditions can improve return to work and work functioning.^{8,50}

The Institute for Work and Health, an independent nonprofit organization, has established seven principles for successful return to work, an evidence-based summary of the most effective workplace interventions, which is intended for all workplaces.²³ The seven principles include (1) a workplace with a strong commitment to health and safety demonstrated by behavior, (2) offer of modified work to injured/ill workers, (3) return-to-work planners

to ensure the plan, (4) supervisors trained in return to work, (5) employers who make early and considerate contact with injured/ill workers, (6) employers and health care providers who communicate with each other, and (7) the worker who has access to clinical interventions for the management of pain, disability, depression, and poor expectations for recovery.⁷

Work Hardening and Work Conditioning

When an employer does not have options for graded activity or modified work, other strategies for return to work are work conditioning, work hardening programs, and functional restoration programs (FRPs). PM&R physicians can recommend these programs when there is evidence that the patient has plateaued with routine active physical therapy treatments, and there are physical deficits that the patient needs to meet job demands. Work conditioning is an intensive course of physical therapy for the purpose of training toward a specific job requirement, such as repetitive tasks or increasing capacities for reaching, standing, walking, or lifting. For health care providers who are recommending work conditioning, it is important to document that the patient has undergone physical therapy with benefits but has plateaued and has not been able to meet the work requirements mandated by the current position. Work conditioning can help with work-specific duties by focusing on building endurance.

Work hardening, which is more intensive than work conditioning, is an individualized, interdisciplinary, job-specific program to assist the injured worker to return to work. Work hardening is recommended when there is a physical, functional, behavioral, and/or vocational deficit that prevents the ability to achieve job demands. Work hardening uses real or simulated work activities and graded increase in activity to help restore function and vocational ability physically, socially, and psychologically. Participation in a work conditioning or hardening program does not preclude concurrently being at work.

Functional Restoration Programs

An FRP, unlike work conditioning and hardening programs, is an interdisciplinary treatment approach for individuals with disabling musculoskeletal conditions. The programs use the biopsychosocial model of rehabilitation, focusing on active conditioning activities, self-directed exercises, and psychosocial interventions using a team approach with a combination of medical doctors, physical therapists, occupational therapists, psychologists, and vocational therapists. The focus and emphasis are on functional improvement rather than elimination of pain. An FRP has shown to decrease sick leave with full-time return-to-work rates ranging from 57% to 80%.^{6,26,44}

The physical demand level at discharge predicted return to work and work retention 1 year after completing an FRP.¹⁶ Furthermore, the length of disability does not decrease the benefits for returning to work or health care utilization. However, the use of an FRP early in the rehabilitative process is more cost effective when compared to receiving the treatment after the individual has exhausted all other treatments and as a treatment of last resort.⁵¹ The ODG state there are several criteria for admission into a functional restoration or chronic pain program. The criteria state that the patient has a chronic pain condition with loss of functional capacity that lasts longer than 3 months, that the patient has tried first-line treatment that has been unsuccessful, and that the claimant has undergone a thorough multidisciplinary evaluation, which usually includes an evaluation by a medical

doctor, physical or occupational therapist, and pain psychologist. Once the multidisciplinary evaluation has taken place, the specific functional goals of the program are individualized to each patient. As the patient undergoes treatment, compliance with the program and subjective and objective functional gains are documented. An FRP is not to exceed 160 hours.

Functional Capacity Evaluation

The FCE is a multifaceted process designed to objectively assess an individual's functional capacity for a return-to-work assessment. FCEs exist for both job-specific functional assessment and overall assessment.²¹ FCEs are recommended prior to and upon completion of an FRP, work conditioning, or work hardening program directed toward a specific job. FCEs can be valuable for determining return to work for a job with specific goals and requirements.³¹ A group of 354 patients with chronic disabling musculoskeletal disorders underwent FCEs at both admission and discharge from their FRP. Among them, 96% improved their physical demand levels and 56% met the lifting requirements of their job, with strength gains approaching 50%. Improvement in FCE-determined physical demand was predictive of returning to work upon discharge and retaining employment 1 year later.¹⁶

An FCE is more likely to produce meaningful results when the worker is properly motivated, collaborates with the examiner, and the testing has a specific, purposeful focus. Providing the evaluator with a job description from the employer enhances the FCE by allowing for a more accurate assessment of the worker's abilities and capacities. Because an FCE evaluates a person's work capacity to determine fitness for duty, it should not be used solely to assess a worker's effort or compliance, assign permanent work restrictions, determine further treatment, predict reinjury, or establish MMI.¹⁷

There continues to be a lack of evidence regarding the reliability and validity of the FCE and how performance correlates to real-life job tasks.¹⁰ Furthermore, there continues to be concerns with inconsistent measurements of endurance and effort. Therefore more research is needed to increase the reliability and validity of the FCE.⁹

Summary

In light of the many individual and societal benefits of maintaining a high employability rate, it is important to effectively implement strategies offered by occupational medicine and vocational rehabilitation teams. A primary focus should be prevention of work-related illnesses, injuries, and fatalities through thoughtful leadership and planning, protective policies, ergonomic design, and a health-conscious workforce. When preexisting or acquired injuries cause limitations in job activities, the multidisciplinary occupational medicine and vocational rehabilitative specialists can be called upon to help treat, compensate, restore, or optimize work participation and function.

Key References

4. Black O, Keegel T, Sim M, et al. The effect of self-efficacy on return-to-work outcomes for workers with psychological or upper-body musculoskeletal injuries: a review of the literature. *J Occup Rehabil.* 2017;28(1):16-27.
5. Blas A, Beltran K, Martinez P, et al. Enabling work: occupational therapy interventions for persons with occupational injuries and diseases: a scoping review. *J Occup Rehabil.* 2017;28(2):201-214.
7. Cancelliere C, Donovan J, Stochkendahl M, et al. Factors affecting return to work after injury or illness: best evidence synthesis of systematic reviews. *Chiropr Man Therap.* 2016;24(1):32.
8. Cullen K, Irvin E, Collie A, et al. Effectiveness of workplace interventions in return-to work for musculoskeletal, pain-related and mental health conditions: an update of the evidence and messages for practitioners. *J Occup Rehabil.* 2017;28(1):1-15.
9. De Baets S, Calders P, Schalley N, et al. Updating the evidence on functional capacity evaluation methods: a systematic review. *J Occup Rehabil.* 2017;28(3):418-428.
10. Denteneer L, Van Daele U, Truijien S, et al. Reliability of physical functioning tests in patients with low back pain: a systematic review. *Spine J.* 2018;18(1):190-207.
13. Escorpizo R, Reneman MF, Ekholm J, et al. A conceptual definition of vocational rehabilitation based on the ICF: building a shared global model. *J Occup Rehabil.* 2011;21(2):126-133.
16. Fore L, Perez Y, Neblett R, et al. Improved functional capacity evaluation performance predicts successful return to work one year after completing a functional restoration rehabilitation program. *PM&R.* 2015;7(4):365-375.
18. Hagen KB, Tambs K, Bjerkedal T. A prospective cohort study of risk factors for disability retirement because of back pain in the general working population. *Spine.* 2002;27(16):1790-1796.
20. Harris-Adamson C, Eisen EA, Kapellusch J, et al. Biomechanical risk factors for carpal tunnel syndrome: a pooled study of 2474 workers. *Occup Environ Med.* 2015;72(1):33-41.
23. Institute for Work and Health. *Seven Principles for Successful Return to Work 2014.* Accessed January 13, 2019. https://www.iwh.on.ca/sites/iwh/files/iwh/tools/seven_principles_rtw_2014.pdf
29. Kamper SJ, Apeldoorn AT, Chiarotto A, et al. Multidisciplinary biopsychosocial rehabilitation for chronic low back pain: Cochrane systematic review and meta-analysis. *BMJ.* 2015;350(5):H444.
34. National Center for Health Statistics, National Vital Statistics System, Centers for Disease Control and Prevention. *Accidents or Unintentional Injuries.* Accessed June 2, 2024. <https://www.cdc.gov/nchs/fastats/accidental-injury.htm>
37. National Institute for Occupational Safety and Health (NIOSH), Centers for Disease Control and Prevention. *Hierarchy of Controls 2015.* Accessed January 13, 2019. <https://www.cdc.gov/niosh/topics/hierarchy/default.html>
40. Occupational Safety and Health Administration (OSHA). *Ergonomic Workstation eTool.* Accessed January 13, 2019. https://www.osha.gov/SLTC/etools/computerworkstations/checklist_evaluation.html
41. Occupational Safety and Health Administration (OSHA). *Injury and Illness Prevention Programs* [White Paper]. January 2012. US Department of Labor. Accessed November 17, 2018. <https://www.osha.gov/dsg/InjuryIllnessPreventionProgramsWhitePaper.html>
47. Rothstein MA. Innovations of the Americans with Disabilities Act: confronting disability discrimination in employment. *JAMA.* 2015;313(22):2221-2222.
51. Theodore B, Mayer T, Gatchel R. Cost-effectiveness of early versus delayed functional restoration for chronic disabling occupational musculoskeletal disorders. *J Occup Rehabil.* 2014;25(2):303-315.
54. US Bureau of Labor Statistics. *Census of Fatal Occupational Injuries (CFOI) Summary*, 2022. December 19, 2023. Accessed June 2, 2024. <https://stats.bls.gov/news.release/cfoi.nr0.htm>
59. van Vilsteren M, van Oostrom S, de Vet H, et al. Workplace interventions to prevent work disability in workers on sick leave. *Cochrane Database Syst Rev.* 2015(10):CD006955.

Visit Elsevier eBooks+ at eBooks.Health.Elsevier.com for a complete list of references.

References

- Andersen JH, Thomsen JF, Overgaard E, et al. Computer use and carpal tunnel syndrome: a 1-year follow-up study. *JAMA*. 2003;289(22):2963-2969.
- Atlas SJ, Chang Y, Kammann E, et al. Long-term disability and return to work among patients who have a herniated lumbar disc: the effect of disability compensation. *J Bone Joint Surg Am*. 2000;82(1):4-15.
- Bigos SJ, Battié MC, Spengler DM, et al. A prospective study of work perceptions and psychosocial factors affecting the report of back injury [published correction appears in *Spine*. 1991;16(6):688]. *Spine*. 1991;16(1):1-6.
- Black O, Keegel T, Sim M, et al. The effect of self-efficacy on return-to-work outcomes for workers with psychological or upper-body musculoskeletal injuries: a review of the literature. *J Occup Rehabil*. 2017;28(1):16-27.
- Blas A, Beltran K, Martinez P, et al. Enabling work: occupational therapy interventions for persons with occupational injuries and diseases: a scoping review. *J Occup Rehabil*. 2017;28(2):201-214.
- Caby I, Olivier N, Janik F, et al. A controlled and retrospective study of 144 chronic low back pain patients to evaluate the effectiveness of an intensive functional restoration program in France. *Healthcare*. 2016;4(2):23.
- Cancelliere C, Donovan J, Stochkendahl M, et al. Factors affecting return to work after injury or illness: best evidence synthesis of systematic reviews. *Chiropr Man Therap*. 2016;24(1):32.
- Cullen K, Irvin E, Collie A, et al. Effectiveness of workplace interventions in return-to-work for musculoskeletal, pain-related and mental health conditions: an update of the evidence and messages for practitioners. *J Occup Rehabil*. 2017;28(1):1-15.
- De Baets S, Calders P, Schalley N, et al. Updating the evidence on functional capacity evaluation methods: a systematic review. *J Occup Rehabil*. 2017;28(3):418-428.
- Denteneer L, Van Daele U, Truijien S, et al. Reliability of physical functioning tests in patients with low back pain: a systematic review. *Spine J*. 2018;18(1):190-207.
- Dworkin RH, Handlin DS, Richlin DM, et al. Unraveling the effects of compensation, litigation, and employment on treatment response in chronic pain. *Pain*. 1985;23(1):49-59.
- Employer Assistance and Resource Network (EARN) on Disability Inclusion. *Job Seeker Resources*. Accessed December 23, 2018. www.askearn.org/resources/job-seeker-resources.
- Escorpizo R, Reneman ME, Ekholm J, et al. A conceptual definition of vocational rehabilitation based on the ICF: building a shared global model. *J Occup Rehabil*. 2011;21(2):126-133.
- Feuerstein M, Miller VL, Burrell LM, et al. Occupational upper extremity disorders in the federal workforce. Prevalence, health care expenditures, and patterns of work disability. *Occup Environ Med*. 1998;40(6):546-555.
- Finger ME, Escorpizo R, Bostan C, et al. Work Rehabilitation Questionnaire (WORQ): development and preliminary psychometric evidence of an ICF-based questionnaire for vocational rehabilitation. *J Occup Rehabil*. 2014;24(3):498-510.
- Fore L, Perez Y, Neblett R, et al. Improved functional capacity evaluation performance predicts successful return to work one year after completing a functional restoration rehabilitation program. *PM&R*. 2015;7(4):365-375.
- Gelfman R, Hill 3rd JJ. Rehabilitating the injured/ill worker to maximum medical improvement (MMI). *Phys Med Rehabil Clin N Am*. 2019;30(3):657-669.
- Hagen KB, Tambs K, Bjerkedal T. A prospective cohort study of risk factors for disability retirement because of back pain in the general working population. *Spine*. 2002;27(16):1790-1796.
- Harbin G, Olson J. Post-offer, pre-placement testing in industry. *Am J Ind Med*. 2005;47(4):296-307.
- Harris-Adamson C, Eisen EA, Kapellusch J, et al. Biomechanical risk factors for carpal tunnel syndrome: a pooled study of 2474 workers. *Occup Environ Med*. 2015;72(1):33-41.
- Harten J. Functional capacity evaluation. *Occup Med*. 1998;13(1):209-212.
- Hoy D, March L, Brooks P, et al. The global burden of low back pain: estimates from the Global Burden of Disease 2010 study. *Ann Rheum Dis*. 2014;73(6):968-974.
- Institute for Work and Health. *Seven Principles for Successful Return to Work 2014*. Accessed January 13, 2019. https://www.iwh.on.ca/sites/iwh/files/iwh/tools/seven_principles_rtw_2014.pdf
- International Labour Organization. *Decent Work*. Accessed October 18, 2018. <https://www.ilo.org/global/topics/decent-work/lang-en/index.htm>
- Jin RL, Shah CP, Svoboda TJ. The impact of unemployment on health: a review of the evidence. *CMAJ*. 1995;153(5):529-540.
- Jousset N, Fanello S, Bontoux L, et al. Effects of functional restoration versus 3 hours per week physical therapy: a randomized controlled study. *Spine*. 2004;29(5):487-493.
- Kalski L, Völkel L, Häußler S, et al. Efficacy of occupational rehabilitation in return to work for back pain: a systematic literature review. *Work*. 2024;78(1):29-43.
- Kammoun N, Bani M, Nouaigui H. The role of occupational medicine in the response to the coronavirus outbreak: the Tunisian Occupational Health and Safety Institute's experience. *Pan Afr Med J*. 2022;41:19.
- Kamper SJ, Apeldoorn AT, Chiarotto A, et al. Multidisciplinary biopsychosocial rehabilitation for chronic low back pain: Cochrane systematic review and meta-analysis. *BMJ*. 2015;350(5):H444.
- Ladou J, Harrison R. The practice of occupational medicine. In: *Current Occupational and Environmental Medicine*. 5th ed. McGraw-Hill Education; 2014:1-4.
- Lyth J. Disability management and functional capacity evaluations: a dynamic resource. *Work*. 2001;16(1):13-22.
- Molde Hagen E, Grasdal A, Eriksen HR. Does early intervention with a light mobilization program reduce long-term sick leave for low back pain: a 3-year follow-up study. *Spine*. 2003;28(20):2309-2315.
- Nathan PA, Meadows KD, Istvan JA. Predictors of carpal tunnel syndrome: an 11-year study of industrial workers. *J Hand Surg Am*. 2002;27(4):644-651.
- National Center for Health Statistics, National Vital Statistics System, Centers for Disease Control and Prevention. *Accidents or Unintentional Injuries*. Accessed June 2, 2024. <https://www.cdc.gov/nchs/fastats/accidental-injury.htm>
- National Industries for the Blind. *Welcome to NIB CareersWith Vision*. Accessed December 23, 2018. www.afb.org/careerconnect/careerswithvision.asp
- National Institute for Occupational Safety and Health (NIOSH), Centers for Disease Control and Prevention. *Elements of Ergonomic Programs 1997*. Accessed January 13, 2019. <https://www.cdc.gov/niosh/topics/ergonomics/ergoprimer/default.html>
- National Institute for Occupational Safety and Health (NIOSH), Centers for Disease Control and Prevention. *Hierarchy of Controls 2015*. Accessed January 13, 2019. <https://www.cdc.gov/niosh/topics/hierarchy/default.html>
- National Safety Council, Work Overview. *Top Work-Injury Related Causes*. Accessed July 8, 2024. <https://injuryfacts.nsc.org/work/work-overview/top-work-related-injury-causes/>
- Nylén L, Voss M, Floderus B. Mortality among women and men relative to unemployment, part time work, overtime work, and extra work: a study based on data from the Swedish twin registry. *Occup Environ Med*. 2001;58(1):52-57.
- Occupational Safety and Health Administration (OSHA). *Ergonomic Workstation eTool*. Accessed January 13, 2019. https://www.osha.gov/SLTC/etools/computerworkstations/checklist_evaluation.html
- Occupational Safety and Health Administration (OSHA). *Injury and Illness Prevention Programs* [White Paper]. January 2012. US Department of Labor. Accessed November 17, 2018. <https://www.osha.gov/dsg/InjuryIllnessPreventionProgramsWhitePaper.html>

42. Oleske DM, Neelakantan J, Andersson GB, et al. Factors affecting recovery from work-related, low back disorders in autoworkers. *Arch Phys Med Rehabil.* 2004;85(8):1362-1364.
43. Pomerance J. Return to work in the setting of upper extremity illness. *J Hand Surg Am.* 2009;34(1):137-141.
44. ReEmployAbility. *Transition2Work.* Accessed December 23, 2018. www.reemployability.com/Transition-2-Work
45. Rempel DM, Janowitz IL. Ergonomics and the prevention of occupational injuries. In: *Current Occupational and Environmental Medicine*. 5th ed. McGraw-Hill Education; 2014:197-222.
46. Roche-Leboucher G, Petit-Lemanach A, Bontoux L, et al. Multidisciplinary intensive functional restoration versus outpatient active physiotherapy in chronic low back pain. *Spine.* 2011;36(26):2235-2242.
47. Rothstein MA. Innovations of the Americans with Disabilities Act: confronting disability discrimination in employment. *JAMA.* 2015;313(22):2221-2222.
48. Schmidt SH, Oort-Marburger D, Meijman TF. Employment after rehabilitation for musculoskeletal impairments: the impact of vocational rehabilitation and working on a trial basis. *Arch Phys Med Rehabil.* 1995;76(10):950-954.
49. Social Security Administration. *Annual Statistical Supplement, 2017. Workers' Compensation Program Description and Legislative History.* Accessed July 8, 2024. <https://www.ssa.gov/policy/docs/statcomps/supplement/2017/workerscomp.html#:~:text=Nine%20states%20enacted%20workers'%20compensation,Islands%20have%20their%20own%20program>
50. Staal J, Hlobil H, Twisk J, et al. Graded activity for low back pain in occupational health care. *Ann Intern Med.* 2004;140(2):77.
51. Theodore B, Mayer T, Gatchel R. Cost-effectiveness of early versus delayed functional restoration for chronic disabling occupational musculoskeletal disorders. *J Occup Rehabil.* 2014;25(2):303-315.
52. US Department of Labor. *Regulations Implementing Section 503 of the Rehabilitation Act 2014.* Accessed December 23, 2018. www.dol.gov/ohccp/regs/compliance/section503.htm
53. US Department of Labor. *Return to Work Toolkit: For Employee and Employers.* Accessed December 31, 2018. <https://www.dol.gov/odep/return-to-work/employer-background/htm>
54. US Bureau of Labor Statistics. *Census of Fatal Occupational Injuries (CFOI) Summary, 2022.* December 19, 2023. Accessed June 2, 2024. <https://stats.bls.gov/news.release/cfoi.nr0.htm>
55. US Bureau of Labor Statistics. *Employer-Reported Workplace Injuries and Illnesses—2021–2022.* News Release November 9, 2023. Accessed June 2, 2024. <https://www.bls.gov/news.release/osh.nr0.htm>
56. US Department of Veterans Affairs. *Vocational Rehabilitation and Employment (VR&E) Longitudinal Study (PL 110-389 Sec. 334): Annual Report 2018 for FY 2017.* <https://www.benefits.va.gov/VOCREHAB/docs/FY2022-longitudinal-study.pdf>
57. US Equal Employment Opportunity Commission. *The Americans with Disabilities Act: A Primer for Small Business.* Accessed July 9, 2024. www.eeoc.gov/eeoc/publications/adahandbook.cfm
58. USAJOBS. *Working in Government: Individuals with Disabilities.* Accessed December 23, 2018. www.usajobs.gov/Help/working-in-government/unique-hiring-paths/individuals-with-disabilities
59. van Vilsteren M, van Oostrom S, de Vet H, et al. Workplace interventions to prevent work disability in workers on sick leave. *Cochrane Database Syst Rev.* 2015(10):CD006955.
60. Wong J, Kallish N, Crown D, et al. Job accommodations, return to work and job retention of people with physical disabilities: a systematic review. *J Occup Rehabil.* 2021;31(3):474-490.
61. World Health Organization, World Bank. *World Report on Disability.* WHO; 2011.
62. World Health Organization. *Classifications.* Accessed December 30, 2018. <https://www.who.int/classifications/icf/en/>